

Post Lab 01 Tasks

You are working as a network administrator. A user reports that they cannot access any device on the network.

Before troubleshooting further, you need to check the user's current IP address and network configuration.

Task # 01

Which command will you use on a Windows computer to check the IP address and network configuration? > **ipconfig**

A user provides the IP address of a server but you need to verify the **hostname of your current computer** before continuing your investigation.

Task # 02

Which command will you use to find the hostname? > **hostname**

```
PS C:\Users\alyxb> hostname
pc2022
PS C:\Users\alyxb> |
```

You are a network administrator. A user reports:

"I can access websites using their IP addresses, but I cannot access them using domain names."

You suspect a **DNS-related problem**.

Task # 03

Which command can you use to investigate whether DNS is resolving the domain name correctly? > **nslookup** www.example.com

```
PS C:\Users\alyxb> nslookup www.example.com
Server:      UnKnown
Address:     fe80::1

Non-authoritative answer:
Name:        www.example.com
Addresses:   2606:4700:10::6814:179a
             2606:4700:10::ac42:93f3
             104.20.23.154
             172.66.147.243
```

A user can communicate with other computers in the same LAN but cannot access networks outside the LAN.

You suspect that the **default gateway** may be unreachable.

The gateway IP is **192.168.1.1**.

Task # 04

Which command will you use to test connectivity to the default gateway? > **ping** 192.168.1.1

```
PS C:\Users\alyxb> ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=2ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

Users report that accessing the company's remote server **10.10.20.50** takes several seconds. You need to determine whether the delay is occurring somewhere along the network path.

Task # 05

Which command should you use on Windows? What information will you analyze in its output?

> **tracert 10.10.20.50 | which node is causing the blockage/slowdown**

You are working as a system administrator investigating a server. Users report that a network application is not working.

You want to check the server's current **network connections and listening ports**.

Task # 06

Which command from the lab manual can you use? What information should you look for in its output? > **netstat | check if the server is listening and check whether its using the correct protocol (TCP or UDP), check whether the port its listening on is free.**

IP Addresses and OSI Model

The private **Class B** IP range is **128.0.0.0** to **191.255.255.255**.

The default subnet mask for **Class A** is **255.0.0.0**

IP Address: 172.20.5.100 What is its default subnet mask? > **255.0.0.0**

Write down three application layer protocols. > **SMTP, POP3, IMAP**

255.0.0.0 default subnet mask is linked to class **A**.

Write down the range of class C IP addresses **192.0.0.0** to **223.255.255.255**